

```
In [1]: # Gain intuition by demonstrating the output of the tangential hyperbolic function over a range of input values.

# tanh(x) = (e^x - e^-x) / (e^x + e^-x)

# The output of tanh(x) is centered around zero and approaches -1 and 1 asymptotically.

# tanh(x) is used as an activation fuction in Recurrent Neural Networks, i.e. neural networks designed
# to process sequence data, such as stock prices or sequences of words, where order is significant.

# The zero-centered output of tanh(x) helps networks converge and learn faster.

# Daniel R. South
# 2026-06-13
```

```
In [2]: import numpy as np
import matplotlib.pyplot as plt

print("Generate a list of x values from -5 to 5")
x1 = np.linspace(-5, 5, 21)
print(x1)

y1 = np.tanh(x1)
print(y1)

print("\nGenerate a list of x values from -2 to 2")
x2 = np.linspace(-2, 2, 21)
print(x2)

y2 = np.tanh(x2)
print(y2)

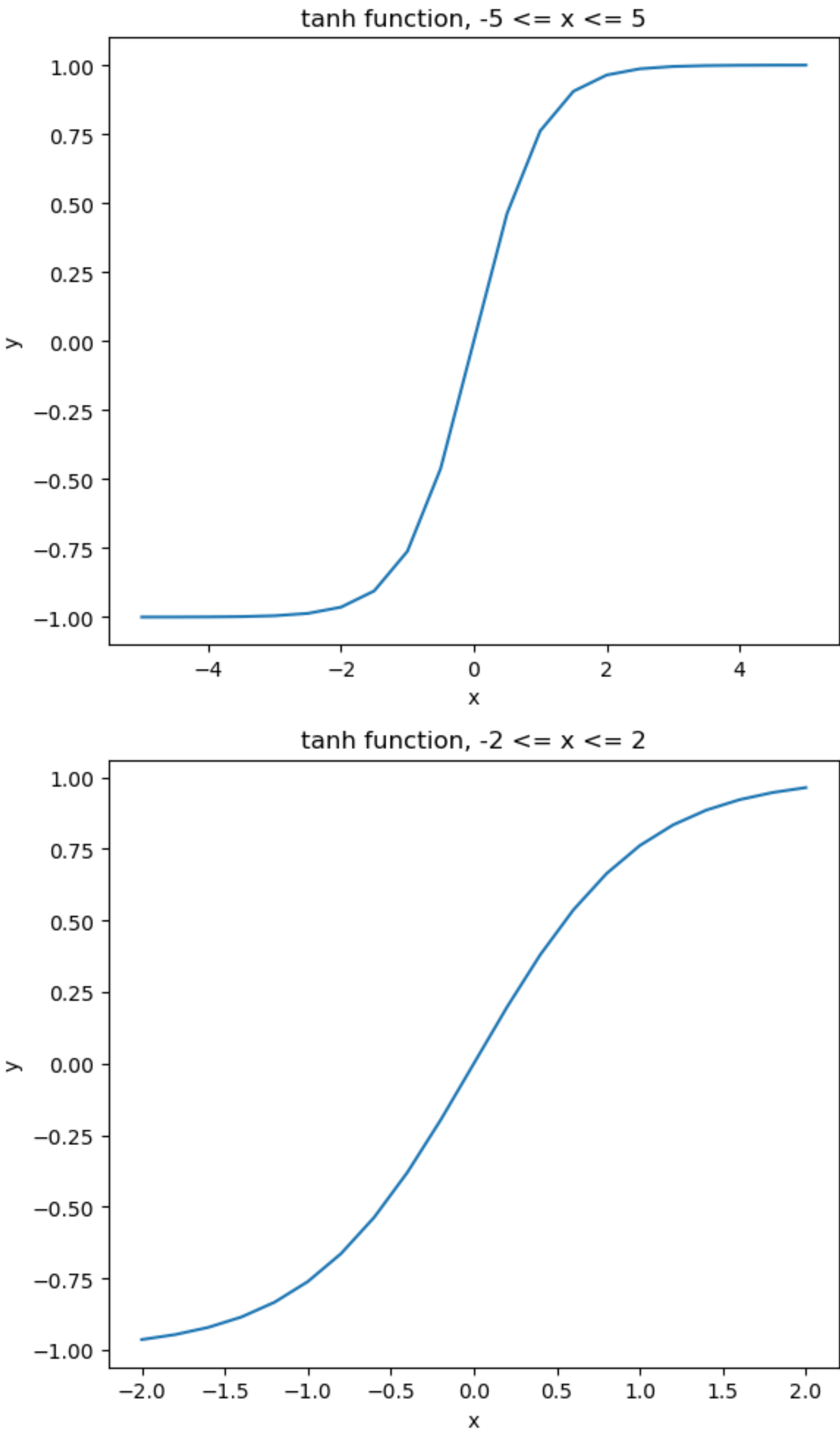
Generate a list of x values from -5 to 5
[-5.  -4.5 -4.   -3.5 -3.   -2.5 -2.   -1.5 -1.   -0.5  0.    0.5  1.    1.5
  2.    2.5  3.    3.5  4.    4.5  5. ]
[-0.9999092 -0.99975321 -0.9993293 -0.9981779 -0.99505475 -0.9866143
 -0.96402758 -0.90514825 -0.76159416 -0.46211716  0.         0.46211716
  0.76159416  0.90514825  0.96402758  0.9866143  0.99505475  0.9981779
  0.9993293  0.99975321  0.9999092 ]

Generate a list of x values from -2 to 2
[-2.  -1.8 -1.6 -1.4 -1.2 -1.   -0.8 -0.6 -0.4 -0.2  0.    0.2  0.4  0.6
  0.8  1.   1.2  1.4  1.6  1.8  2. ]
[-0.96402758 -0.94680601 -0.92166855 -0.88535165 -0.83365461 -0.76159416
 -0.66403677 -0.53704957 -0.37994896 -0.19737532  0.         0.19737532
  0.37994896  0.53704957  0.66403677  0.76159416  0.83365461  0.88535165
  0.92166855  0.94680601  0.96402758]
```

```
In [3]: fig, (ax1, ax2) = plt.subplots(nrows=2, ncols=1, figsize=(6,10))
ax1.plot(x1, y1)
ax1.set_title("tanh function, -5 <= x <= 5")
ax1.set_xlabel("x")
ax1.set_ylabel("y")

ax2.plot(x2, y2)
ax2.set_title("tanh function, -2 <= x <= 2")
ax2.set_xlabel("x")
ax2.set_ylabel("y")

plt.tight_layout()
plt.show()
```



In []: